NST2AS NATURAL SCIENCES TRIPOS Part II

Monday 3 June 2024 1:30pm – 4:30pm

ASTROPHYSICS - PAPER 1

Before you begin read these instructions carefully.

Candidates may attempt not more than 6 questions.

Each question is divided into Part **(i)** *and Part* **(ii)**, *which may or may not be related. Candidates may attempt either or both parts. The number of marks for each question is the same, with Part* **(ii)** *of each question carrying twice as many marks as Part* **(i)**. *The approximate number of marks allocated to each component of a question is indicated in the right margin. Additional credit will be given for a substantially complete answer to either part.*

Write on **one side** *of the paper only and* **begin each answer on a separate sheet.** *Each sheet should contain the number, letter and part of the question being attempted, and a page number count for this question.*

Answers must be tied up in separate bundles, marked **X, Y, Z**, *according to the letter associated with each question, and a cover sheet must be completed and attached to each bundle. (For example,* **1X, 5X and 8X** *should be in one bundle and* **2Y, 4Y and 7Y** *in another bundle.)*

A master cover sheet listing all parts of all questions attempted must also be completed.

It is essential that every cover sheet bear the candidate's examination number and desk number.

STATIONERY REQUIREMENTS	SPECIAL REQUIREMENTS
Script Paper	*Formulae Booklet*
Blue Cover Sheets	*Approved Calculators Allowed*
Yellow Master Cover Sheets	
1 Rough Work Pad	
Tags	

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the Invigilator

Question 1X – Relativity

(i) The Gödel universe is a solution to the Einstein field equations for an infinitely long rotating dust cylinder. Gödel's metric can be written in terms of dimensionless time t and cylindrical coordinates (r, ϕ, z) as

$$ds^2 = 4a^2 \left(dt^2 - dz^2 - dr^2 + \sinh^2 r(\sinh^2 r - 1)d\phi^2 - 2\sqrt{2}\sinh^2 r \, d\phi dt \right) ,$$

where $-\infty < t < \infty$, $0 \leq r < \infty$, $-\infty < z < \infty$ and a is a positive constant. The angular coordinate ϕ satisfies $0 \leq \phi < 2\pi$, with $\phi = 0$ and 2π being identified.

(a) Consider a curve $x^\mu(\lambda)$ with λ an affine parameter. The curve is defined by $r = r_0$, $t = t_0$ and $z = z_0$, where r_0, t_0 and z_0 are all constants. Write down the tangent vector to the curve. [1]

(b) Find the condition on r_0 for the curve to be timelike for any value of λ. [3]

(c) Now consider an astronaut travelling along the curve that satisfies this condition and performing one full loop (i.e., arriving back to the coordinate starting point such that ϕ ranges from 0 to 2π). Find the proper time elapsed for the astronaut. [4]

(d) Explain why this represents time travel. [2]

(ii) Gödel's metric can also be written in coordinates (x^0, x^1, x^2, x^3), which are dimensionless, as

$$ds^2 = a^2 \left((dx^0)^2 - (dx^1)^2 + \frac{1}{2} e^{2x^1} (dx^2)^2 - (dx^3)^2 + 2e^{x^1} dx^0 dx^2 \right) ,$$

where a is a positive constant. The inverse metric is given by

$$g^{\mu\nu} = \frac{1}{a^2} \begin{pmatrix} -1 & 0 & 2e^{-x^1} & 0 \\ 0 & -1 & 0 & 0 \\ 2e^{-x^1} & 0 & -2e^{-2x^1} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} .$$

(a) Find for which values of μ, ν and σ the partial derivatives $\partial g_{\mu\nu}/\partial x^\sigma$ do not vanish. [1]

QUESTION CONTINUES ON NEXT PAGE

(b) The non-vanishing independent components of the metric connection are

$$\Gamma^0_{12} = \Gamma^1_{02} = \frac{1}{2}e^{x^1}\,, \qquad \Gamma^0_{01} = 1\,, \qquad \Gamma^1_{22} = \frac{1}{2}e^{2x^1}\,, \qquad \Gamma^2_{01} = -e^{-x^1}\,.$$

Verify the result for Γ^0_{12}. [3]

(c) Show that the non-vanishing components of the Ricci tensor are [4]

$$R_{00} = -1\,, \qquad R_{02} = R_{20} = -e^{x^1}\,, \qquad R_{22} = -e^{2x^1}\,.$$

(d) Show that the Ricci scalar $R = -1/a^2$. [2]

(e) Express the Ricci tensor $R_{\mu\nu}$ as an outer product of u_μ with itself, where u^μ is a unit vector in the x^0-direction. [4]

(f) Assuming a universe with a cosmological constant term Λ and filled with dust with energy density ρ, find the values of ρ and Λ as a function of a for which $R_{\mu\nu}$ satisfies the Einstein field equations. [You may work in units with $c = 1$ and may assume that the dust 4-velocity is u^μ.] [5]

(g) Comment on the role of Λ in this solution. [1]

Question 2Y – Astrophysical Fluid Dynamics

(i) (a) In massive stars, radiation pressure becomes significant, exerting an outward force that counteracts gravity. The Eddington limit $L_{\rm Edd}$ is defined as the maximum luminosity L for which radiation pressure does not exceed the inward gravitational pull. Consider an outward radiation acceleration term,

$$\mathbf{g}_{\rm rad} = \frac{\kappa L}{4\pi c r^3}\mathbf{r}\,,$$

where κ is the gas opacity, and derive the Eddington limit at the stellar surface as function of stellar mass M and physical constants. [2]

(b) One of the most massive stars known is R136a1 in the Large Magellanic Cloud galaxy, with a measured luminosity $L = 4.7 \times 10^6\, L_\odot$ and an estimated mass $M = 196\, M_\odot$. Assuming an opacity $\kappa = 0.04\,\mathrm{m^2\,kg^{-1}}$, calculate the ratio between the luminosity of R136a1 and its Eddington luminosity. [2]

(c) The opacity κ generally depends on the gas temperature and composition. It is possible for a star below its Eddington luminosity to have an opacity peak in its interior where radiative acceleration overcomes gravity. Write the equation of hydrostatic equilibrium for a spherically symmetric star with the addition of an outward radiative acceleration term $\mathbf{g}_{\rm rad}$, and derive a condition for hydrostatic equilibrium. [You may assume that the region considered is outside the energy generation region so that L is constant.] [3]

(d) Provide a physical interpretation of the hydrostatic equilibrium condition including radiation pressure in the stellar interior. [3]

(ii) (a) Consider a steady-state, spherically symmetric, isothermal outflow model for a stellar wind that includes the radiative acceleration from Part (i). Neglecting the self-gravity of the gas in the wind, and adopting a barotropic equation of state, show that

$$\left(u^2 - c_{\rm s}^2\right)\frac{d}{dr}\ln u = \frac{2c_{\rm s}^2}{r}\left[1 - \frac{GM}{2c_{\rm s}^2 r}\left(1 - \frac{\kappa L}{4\pi c G M}\right)\right]\,,$$

where $u(r)$ is the outward-directed velocity of the flow, and $c_{\rm s}$ is the sound speed. [7]

(b) Assuming that the outflow is launched from close to the star with a velocity that is significantly subsonic, sketch the possible solution curves for this problem on the $(r/r_{\rm s}, u/c_{\rm s})$-plane, where [3]

$$r_{\rm s} = \frac{GM}{2c_{\rm s}^2}\left(1 - \frac{\kappa L}{4\pi c G M}\right)\,.$$

QUESTION CONTINUES ON NEXT PAGE

(c) Show that in the case that the solution makes a transition from a subsonic to a supersonic flow, [7]

$$\left(\frac{u}{c_\mathrm{s}}\right)^2 - 2\ln\left(\frac{u}{c_\mathrm{s}}\right) = 4\ln\left(\frac{r}{r_\mathrm{s}}\right) + \frac{4r_\mathrm{s}}{r} - 3\,.$$

(d) Consider a Wolf–Rayet star with a mass $M = 25\,M_\odot$ and luminosity $L = 10^6\,L_\odot$, whose wind is composed of fully ionised helium, so that $\kappa = 0.02\,\mathrm{m^2\,kg^{-1}}$. Adopting a temperature of $T = 2\times10^5\,\mathrm{K}$, calculate the location of the sonic point. What is the density at the sonic point if the mass-loss rate of the star is $\dot{M} = 10^{-4}\,M_\odot\,\mathrm{yr^{-1}}$? [3]

Question 3Z – Introduction to Cosmology

(i) (a) Explain what is meant by the term *particle horizon*. [2]

(b) Show that in a Friedmann–Robertson–Walker universe with scale factor $R(t)$, where $R = 0$ at $t = 0$, the (proper) particle horizon at time t is

$$d_{\rm H}(t) = R(t) \int_0^t \frac{c\, dt'}{R(t')} \,.$$

Find $d_{\rm H}(t)$ for a flat, matter-dominated universe. [4]

(c) Assume that at last scattering the age of the Universe is $t = 3 \times 10^5\,$yr and the redshift is $z = 1100$. Estimate the proper size of the particle horizon (in pc) at the epoch of last scattering and calculate the angular size on the sky today subtended by this distance. [You may assume a flat, matter-dominated universe for all relevant epochs.] [4]

(ii) (a) Show that in a Friedmann–Robertson–Walker universe with scale factor $R(t)$, Hubble parameter $H(t)$ and curvature constant K, the density parameter Ω satisfies [3]

$$|\Omega - 1| = \frac{|K|\, c^2}{H^2\, R^2} \,.$$

[You should assume a vanishing cosmological constant.]

(b) Find the dependence of H^2R^2 on R at early times for the two cases of a universe dominated by either radiation or matter and use your results to outline the flatness problem. [3]

(c) Henceforth, consider a universe in which the energy density ρc^2 is dominated by a substance with pressure P and equation of state $P = w\rho c^2$, where w is a constant. Establish a condition on w for the universe to be accelerating. [2]

(d) Explain what is meant by *inflationary expansion*. [2]

(e) Show that the time derivative of the density parameter satisfies

$$\frac{d\Omega}{dt} = H\Omega(\Omega - 1)(1 + 3w) \,,$$

and use this to explain how a period of early inflation solves the flatness problem. [10]

Question 4Y – Structure and Evolution of Stars

(i) An astronomer measures the following positions of a bright single star at yearly intervals.

Date	RA	Dec
1 Jan 2000	12h 30m 15s.1	$+51^\circ\,20'\,15''.0$
1 Jan 2001	12h 30m 16s.9	$+51^\circ\,20'\,32''.3$
1 Jan 2002	12h 30m 18s.0	$+51^\circ\,20'\,42''.5$
1 Jan 2003	12h 30m 18s.2	$+51^\circ\,20'\,44''.5$
1 Jan 2004	12h 30m 19s.0	$+51^\circ\,20'\,52''.1$
1 Jan 2005	12h 30m 20s.4	$+51^\circ\,21'\,05''.7$

(a) From these data, derive the proper motion of the star. [7]

(b) What can you conclude regarding the accuracy of the positional measurements? [3]

(ii) (a) Explain what is meant by radiative opacity κ_ν, and provide a definition of it in terms of the interaction cross-section σ_ν and the number density n of targets. Relate this to the photon mean free path μ_ν. [4]

(b) Briefly describe the sources of radiative opacity in main-sequence stars. State which are the predominant sources of opacity in stars of spectral types O and M. [10]

(c) In the centre of cool white dwarfs composed of pure carbon, detailed calculations show that the *conduction* opacity can be described as

$$\kappa_{\rm cond} \approx \Lambda \left(\frac{T}{\rho}\right)^2 ,$$

where T is the temperature, ρ is the density and $\Lambda = 5 \times 10^{-7}\,{\rm g\,K^{-2}\,cm^{-4}}$ is a constant. Assuming that radiative opacity is dominated by Thomson scattering, show that in the centre of such stars, where $T \approx 10^7\,{\rm K}$ and $\rho \approx 10^6\,{\rm g\,cm^{-3}}$, conduction dominates the energy transport. [6]

Question 5X – Statistical Physics

(i) (a) What systems are described by a *canonical ensemble*? If the energy of microstate n is E_n, with $n = 0, 1, 2, \ldots$, write down an expression for the partition function Z in terms of temperature T and the Boltzmann constant $k_{\rm B}$. [4]

(b) Calculate the partition function Z_1 for a single classical ultra-relativistic spinless particle moving in three-dimensional space in a potential $U(\mathbf{x})$. Here, ultra-relativistic means that the energy-momentum relation is $E = pc$, where c is the speed of light. [6]

(ii) A system of a large number N of non-interacting classical ultra-relativistic spinless particles is in equilibrium at temperature T in a potential

$$U(\mathbf{x}) = \frac{(x^2 + y^2 + z^2)^n}{V^{2n/3}} \,,$$

where n is a positive integer and V is an external parameter analogous to volume.

(a) Calculate the partition function and hence show that the Helmholtz free energy is

$$F = -N k_{\rm B} T \Big[\ln V + A \ln(k_{\rm B} T) + \ln I_n + B \Big] \,,$$

where

$$I_n = \int_0^\infty du\, u^2 \exp\left(-u^{2n}\right) \,,$$

and you should determine A and B. [5]

(b) Considering the conjugate pressure to V, $P = -\left(\frac{\partial F}{\partial V}\right)_{T,N}$, derive the equation of state. [2]

(c) Compute the average energy E, the variance of the energy $(\Delta E)^2$ and heat capacity C_V for the system. Comment on $(\Delta E)/E$ in the thermodynamic limit. [7]

(d) Determine the local particle number density as a function of $\mathbf{x}$ and find the most likely $|\mathbf{x}|$ to find a particle (for $|\mathbf{x}| < \infty$). [6]

Question 6Z – Principles of Quantum Mechanics

(i) (a) A two-dimensional Hilbert space is spanned by two normalised vectors $|\phi\rangle$ and $|\psi\rangle$, not necessarily orthogonal. Consider the linear operator H defined by

$$H\,|\psi\rangle = g\,|\phi\rangle \ , \qquad H\,|\phi\rangle = g^*\,|\psi\rangle \ ,$$

where g is a complex constant. Determine the condition on $|\phi\rangle$, $|\psi\rangle$ and g under which H is Hermitian. [4]

(b) Henceforth, assume this Hermitian condition is satisfied. Determine the eigenvalues of H and the corresponding (un-normalised) eigenvectors $|\pm\rangle$ in the form $|\pm\rangle = |\phi\rangle \pm \gamma\,|\psi\rangle$, where you should specify γ. [6]

(ii) (a) For the two-dimensional system in Part (i), assume now that $g = 1$ in suitable units. The Hamiltonian is given by $H + \Delta(t)$ where

$$\Delta(t)\,|\phi\rangle = \Delta(t)\,|\psi\rangle = V(t)\,(|\phi\rangle + |\psi\rangle) \ ,$$

with $V(t)$ a real and time-dependent function. Working in the $|\pm\rangle$ basis, determine the general solution of the time-dependent Schrödinger equation. [12]

(b) Determine the exact probability to find the system in state $|\phi\rangle$ at time $t > 0$ if it was in state $|\psi\rangle$ at $t = 0$. [8]

Question 7Y – Stellar Dynamics and the Structure of Galaxies

(i) (a) Show that the Laplacian of a spherically symmetric potential $\Phi(r)$ can be written in the form

$$\nabla^2\Phi = \frac{1}{r}\frac{d^2}{dr^2}\left(r\Phi\right) ,$$

where r is the distance from the centre. [2]

(b) A galaxy has the potential of an *isothermal sphere*, namely

$$\Phi(r) = v_0^2 \ln r \,, \qquad (*)$$

where v_0 is a constant. Show that the density distribution that generates this potential is [2]

$$\rho(r) = \frac{v_0^2}{4\pi G r^2} \,. \qquad (\dagger)$$

(c) Derive the velocity of circular orbits in the equatorial plane. [2]

(d) Sketch the circular velocity curve of the isothermal sphere as a function of radius r and compare it to the rotation curves of spiral galaxies, identifying at least one similarity and at least one difference. [4]

(ii) (a) Show that the energy E is an integral of motion of stars moving in the isothermal-sphere potential given in $(*)$ in Part (i). [2]

(b) Explain, without detailed proof, why we may seek a distribution function that is a function of energy, $F = F(E)$. [2]

(c) Using angled brackets to denote averages over the distribution function $F(E)$, demonstrate that the velocity-dispersion tensor has the properties

$$\langle v_r^2 \rangle = \langle v_\theta^2 \rangle = \langle v_\phi^2 \rangle \,,$$

and

$$\langle v_r v_\theta \rangle = \langle v_r v_\phi \rangle = \langle v_\theta v_\phi \rangle = 0 \,,$$

where r, θ and ϕ are spherical polar coordinates. [4]

QUESTION CONTINUES ON NEXT PAGE

(d) By making use of the standard integral (for $\alpha > 0$),

$$\int_{-\infty}^{\infty} \exp\bigl(-\alpha v^2\bigr) dv = \sqrt{\pi/\alpha}\,,$$

verify that the distribution function

$$F(E) = A \exp\left(-2E/v_0^2\right)$$

reproduces the density given by (†) in Part (i), where A is a constant to be found. [4]

(e) Show that the most probable speed of a star is v_0. [2]

(f) Show that the average speed of a star is $2v_0/\sqrt{\pi}$. [2]

(g) Show that the root-mean-square speed of a star is $\sqrt{3/2}v_0$. [2]

(h) What is the escape speed of a star? [2]

Question 8X – Topics in Astrophysics

(i) An observer performs a magnitude-limited spectroscopic survey and identifies a set of 485 stars with spectral features characteristic of solar-type stars. After monitoring of their spectra, it is discovered that 79 of these objects are in fact unresolved binaries composed of two solar-type stars with total luminosity twice that of a single solar-type star.

(a) Estimate the ratio of the number of single solar-type stars to the number of binaries containing two solar-type stars within a fixed volume of space. [8]

(b) Explain carefully the assumptions that you have made in arriving at your answer. [2]

[*You may assume that the absolute magnitude M of a star is given by*

$$M = -2.5 \log_{10}(L/L_\odot) + 4.83\,,$$

and that the apparent magnitude m of a star is related to M and the distance from the observer D by

$$m = M + 5 \log_{10}(D/\mathrm{pc}) - 5\,. \quad]$$

(ii) Photoionisation of neutral hydrogen involves the ejection of an electron as a result of interaction with a photon of ultraviolet light. A star, whose luminosity in the frequency range ν to $\nu + d\nu$ is $L_\nu d\nu$, is located at distance r from a cloud of neutral hydrogen.

(a) Explain why the rate of photoionisation per unit volume for a cloud with a number density of neutral hydrogen atoms n_{neut} is given by

$$\dot{n}_{\mathrm{ion}} = n_{\mathrm{neut}} \int_0^\infty \frac{L_\nu \sigma_\nu}{4\pi r^2 h\nu} d\nu\,,$$

where σ_ν is the photoionisation cross-section per neutral hydrogen atom for photons of frequency ν. [5]

(b) Why is $\sigma_\nu = 0$ for $\nu < \nu_{\mathrm{min}}$, where $h\nu_{\mathrm{min}} = 13.6\,\mathrm{eV}$? [2]

(c) The star's luminosity per unit frequency per unit area of its surface is given by

$$F_\nu = \frac{2\pi h}{c^2} \frac{\nu^3}{\exp\left(\frac{h\nu}{k_{\mathrm{B}}T}\right) - 1}\,.$$

Explain why photoionisation is ineffective if the temperature of the star T_* is less than T_{min}, where $3k_{\mathrm{B}}T_{\mathrm{min}} \approx h\nu_{\mathrm{min}}$. [3]

QUESTION CONTINUES ON NEXT PAGE

(d) For stars with $T_* \gg T_{\rm min}$, find how the rate of photoionisation scales with T_* and stellar radius R_* if $\sigma_\nu \propto \nu^{-3}$ for $\nu > \nu_{\rm min}$. [4]

(e) Given that stellar luminosity and radius scale with mass according to $L_* \propto M_*^{3.5}$ and $R_* \propto M_*^{0.6}$, determine how the rate of photoionisation of a cloud of fixed properties at a fixed distance from the star depends on stellar mass. [3]

(f) Now consider the case where the cloud is located close to a cluster of bright young stars whose mass function is described by the Salpeter law (fraction of stars per unit mass interval scaling as $M_*^{-2.35}$). Determine whether the total ionisation rate is dominated by stars with T only mildly greater than $T_{\rm min}$ or by the hottest stars in the cluster. [3]

END OF PAPER